Homework 4

Grant Talbert
February 20, 2026
MA731

Boston University

Problem 1. *Using the Jordan canonical form, show that every complex square matrix is the limit of a sequence of diagonalizable matrices. (For instance consider the vectors $e_1 + \varepsilon e_2 + \cdots + \varepsilon^{k-1} e_k$.)*

Proof. Let $A \in \mathfrak{gl}(n, \mathbb{C})$. Since $\mathbb{C}$ is algebraically closed, A is conjugate to its Jordan canonical form J, meaning $A = PJP^{-1}$ for some invertible $P \in \mathrm{GL}(n, \mathbb{C})$. Recall that J is given by the block diagonal matrix $J = \mathrm{diag}(J_{k_1}(\lambda_1), \ldots, J_{k_m}(\lambda_m))$, where each Jordan block $J_k(\lambda)$ is of the form

$$J_k(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & \cdots & 0 & \lambda \end{pmatrix}.$$

It suffices to show each Jordan block is a limit of some sequence of diagonalizable matrices, since we can take the direct sum of the sequences to obtain a sequence converging to J, then simply conjugate each term in the sequence.

Let $J_k(\lambda)$ be a Jordan block $\lambda 1 + N$ for N the matrix having all 1s on the first diagonal above the main diagonal, and 0s elsewhere. Note that if $\{e_i\}_{1 \leq i \leq k}$ is the standard basis, then $Ne_j = e_{j-1}$ for $j > 1$, and $Ne_1 = 0$. Let $\varepsilon \neq 0$ and define

$$M(\varepsilon) = J_k(\lambda) + \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \varepsilon^k & 0 & \cdots & 0 \end{pmatrix}.$$

Write $\varepsilon^k E$ for the matrix on the far right, such that E is the matrix with a 1 in the bottom left corner, and 0s elsewhere. Note that

$$Ee_i = \delta_{i1} e_k.$$

Let

$$v(\varepsilon) \coloneqq \sum_{j=1}^{k} \varepsilon^{j-1} e_j = e_1 + \varepsilon e_2 + \cdots + \varepsilon^{k-1} e_k.$$

We compute $M(\varepsilon)v(\varepsilon)$:

$$\begin{aligned}
M(\varepsilon)v(\varepsilon) &= (\lambda 1 + N + \varepsilon^k E)\sum_{j=1}^{k} \varepsilon^{j-1} e_j \\
&= \sum_{j=1}^{k} \lambda \varepsilon^{j-1} e_j + \sum_{j=1}^{k} \varepsilon^{j-1} N e_j + \sum_{j=1}^{k} \varepsilon^{j-1}\varepsilon^k E e_j \\
&= \lambda v(\varepsilon) + \sum_{j=1}^{k-1} \varepsilon^{j-1} e_{j-1} + \varepsilon^k E e_1 \\
&= \lambda v(\varepsilon) + \varepsilon\left(v(\varepsilon) - \varepsilon^{k-1} e_k\right) + \varepsilon^k e_k \\
&= \lambda v(\varepsilon) + \varepsilon\left(v(\varepsilon) - \varepsilon^{k-1} e_k + \varepsilon^{k-1} e_k\right) \\
&= (\lambda + \varepsilon) v(\varepsilon).
\end{aligned}$$

Thus $v(\varepsilon)$ is an eigenvector of $M(\varepsilon)$ with eigenvalue $\lambda + \varepsilon$. We can extend this to a full basis of eigenvectors by taking ω to be a primitive kth root of unity $e^{2i\pi/k}$, and considering $v(\omega^m \varepsilon)$ for any $m \in \mathbb{Z}$. There are of course k distinct vectors of this form, and we can once again compute (since $(\omega^m)^k = (\omega^k)^m = 1$)

$$\begin{aligned}
M(\varepsilon)v(\omega^m\varepsilon) &= (\lambda 1 + N + \varepsilon^k E)\sum_{j=1}^{k} (\omega^m\varepsilon)^{j-1} e_j \\
&= \sum_{j=1}^{k} \lambda (\omega^m\varepsilon)^{j-1} e_j + \sum_{j=1}^{k} (\omega^m\varepsilon)^{j-1} N e_j + \sum_{j=1}^{k} (\omega^m\varepsilon)^{j-1}\varepsilon^k E e_j \\
&= \lambda v(\omega^m\varepsilon) + \sum_{j=1}^{k-1} (\omega^m\varepsilon)^{j-1} e_{j-1} + \varepsilon^k E e_1 \\
&= \lambda v(\omega^m\varepsilon) + \omega^m\varepsilon\left(v(\omega^m\varepsilon) - (\omega^m\varepsilon)^{k-1} e_k\right) + \varepsilon^k e_k \\
&= \lambda v(\omega^m\varepsilon) + \omega^m\varepsilon\left(v(\omega^m\varepsilon) - (\omega^m\varepsilon)^{k-1} e_k + (\omega^m\varepsilon)^{k-1} e_k\right) \\
&= (\lambda + \omega^m\varepsilon) v(\omega^m\varepsilon).
\end{aligned}$$

Because $\varepsilon \neq 0$, these eigenvalues are distinct, and thus $M(\varepsilon)$ is always diagonalizable. Now simply take the sequence $\{\varepsilon_m = 1/m\}_{m\in\mathbb{N}}$ which converges $\varepsilon_m \to 0$ as $m \to \infty$, meaning

$$\lim_{m\to\infty} M(\varepsilon_m) = \lim_{m\to\infty}\left(\lambda 1 + N + \varepsilon_m^k E\right) = \lambda 1 + N = J_k(\lambda).$$

Hence every single Jordan block matrix is a limit of diagonalizable matrices. We can take the direct sum to obtain that J is a limit of diagonalizable matrices. Conjugating each limit term by P yields A, and the conjugate of a diagonalizable matrix is diagonalizable, hence A is a limit of diagonalizable matrices. ■

Problem 2. *Using* $\log(-)$*, show that for* $X \in \mathfrak{gl}(n, \mathbb{C})$*,*

$$\lim_{m\to\infty}\left(1 + \frac{X}{m}\right)^m = e^X.$$

Proof. Recall ([Hal15] def 2.7) that $\log X$ can be written as

$$\log X = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(X-1)^n}{n}.$$

Then

$$\log\left(1 + \frac{X}{m}\right) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{X^n}{nm^n}.$$

We must of course choose X, m such that $\left\|\frac{X}{m}\right\| < 1$ for the power series to converge. In particular, there exists sufficiently large m for which the sequence converges, so the sum is defined in the limit $m \to \infty$. We can multiply by m to write

$$m \log\left(1 + \frac{X}{m}\right) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{X^n}{nm^{n-1}} = X + \sum_{n=2}^{\infty} (-1)^{n+1} \frac{X^n}{nm^{n-1}}.$$

Write $R = \sum_{n=2}^{\infty} (-1)^{n+1} X^n/(nm^{n-2})$ such that

$$m \log\left(1 + \frac{X}{m}\right) = X + \frac{R}{m}.$$

Then

$$\|R\| = \left\| \sum_{n=2}^{\infty} (-1)^{n+1} \frac{X^n}{nm^{n-2}} \right\| \le \sum_{n=2}^{\infty} \frac{\|X\|^n}{nm^{n-2}} \le \sum_{n=2}^{\infty} \frac{\|X\|^n}{m^{n-2}}.$$

We claim that when $m > 1$ and $m > 2\|X\|$, this is a convergent geometric series. This follows by

$$\sum_{n=2}^{\infty} \frac{\|X\|^n}{m^{n-2}} = \|X\|^2 \sum_{n=0}^{\infty} \frac{\|X\|^n}{m^n} = \|X\|^2 \sum_{n=0}^{\infty} \left(\frac{\|X\|}{m}\right)^n,$$

and this converges when $\|X\|/m < 1$. If we take $m > 2\|X\|$, then $\|X\|/m < 1/2$ implies that this converges to a value strictly less than

$$\sum_{n=0}^{\infty} \left(\frac{\|X\|}{m}\right)^n < \frac{1}{1 - \frac{1}{2}} = \frac{1}{\frac{2}{2} - \frac{1}{2}} = 2.$$

Hence $\|R\| < 2\|X\|^2$, and thus R is bounded by a constant for sufficiently large m. Hence the sum converges in the limit $m \to \infty$, and in particular $R/m \to 0$ as $m \to \infty$. Thus

$$\lim_{m\to\infty} m \log\left(1 + \frac{X}{m}\right) = \lim_{m\to\infty} \left(X + \frac{R}{m}\right) = X.$$

Since $\exp \circ \log = \mathrm{id}$, and since $\exp(mA) = (\exp A)^m$ for all $A \in \mathfrak{gl}(n, \mathbb{C})$, we have

$$\exp\left(m \log\left(1 + \frac{X}{m}\right)\right) = \exp\left(\log\left(1 + \frac{X}{m}\right)\right)^m = \left(1 + \frac{X}{m}\right)^m.$$

Since the matrix exponential is everywhere continuous, it commutes with the limit, and thus

$$\exp X = \exp\left(\lim_{m\to\infty} \log\left(1 + \frac{X}{m}\right)\right) = \lim_{m\to\infty} \exp\left(\log\left(1 + \frac{X}{m}\right)\right) = \lim_{m\to\infty} \left(1 + \frac{X}{m}\right)^m.$$

■

References

[Hal15] Brian Hall. *Lie Groups, Lie Algebras, and Representations.* Springer GTM, 2015.